

Invest Immo - Full Analysis Methodology

This document is the authoritative functional specification for how the simulation computes results.

It explains:

- all inputs,
- all formulas,
- all strategy-specific hypotheses,
- all output indicators,
- and the current model limits.

This is a decision-support model, not legal, tax, or accounting advice.

1) Scope

Backend endpoint:

- POST /api/analyse

For each run, the backend returns:

- a global summary block,
- 5 strategy analyses:
 - Location Nue
 - LMNP Meuble
 - Location Courte Duree (Airbnb-like)
 - SCPI
 - Residence Principale

Each strategy contains:

- yearly rows from year 1 to horizon,
- synthesized indicators (cash flow moyen, TRI, rendement brut, patrimoine final).

2) Input Parameters

2.1 Core financing and market inputs

- apport
- taux_interet
- revenu_mensuel_brut
- prix_m2_achat
- loyer_m2
- surface
- duree_pret_ans
- frais_notaire_pct
- charges_copro_mensuelle
- taxe_fonciere_mensuelle

- vacance_locative_pct
- gestion_locative_pct
- tmi
- revalorisation_bien_pct
- revalorisation_loyer_pct
- horizon_ans
- rendement_scp

2.2 LMNP-specific inputs (current model)

- travaux_annuel
- assurance_pno_annuel
- cfe_annuel
- honoraires_comptable_annuel
- amort_frais_acq_annuel

2.3 Multi-flat mode overrides

In the multi-flat front-end flow, each flat can override:

- prix_m2_achat
- loyer_m2
- surface
- frais_notaire_pct
- charges_copro_mensuelle
- taxe_fonciere_mensuelle
- vacance_locative_pct
- gestion_locative_pct

The backend is still called once per flat. Each call is independent.

3) Core Engine Calculations (used by several strategies)

3.1 Acquisition and financing

- $\text{prix_achat} = \text{prix_m2_achat} * \text{surface}$
- $\text{frais_notaire} = \text{prix_achat} * \text{frais_notaire_pct} / 100$
- $\text{cout_total} = \text{prix_achat} + \text{frais_notaire}$
- $\text{emprunt} = \max(0, \text{cout_total} - \text{apport})$

3.2 Loan monthly payment

Let:

- $t = \text{taux_interet} / 100 / 12$
- $n = \text{duree_pret_ans} * 12$

If $\text{emprunt} \leq 0$, $\text{mensualite} = 0$.

If $t == 0$:

- $\text{mensualite} = \text{emprunt} / n$

Else (annuity formula):

- $\text{mensualite} = \text{emprunt} * t / (1 - (1 + t)^{-n})$

3.3 Amortization table

For each month:

- $\text{interets_m} = \text{capital_restant} * t$
- $\text{amort_m} = \text{mensualite} - \text{interets_m}$
- $\text{capital_restant} = \max(0, \text{capital_restant} - \text{amort_m})$

Yearly helpers:

- $\text{interets annuels} = \text{sum of monthly interest over the 12 months of year } y$
- $\text{capital restant (year } y) = \text{remaining principal at the end of year } y$

3.4 Shared tax rate

- $\text{taux_fiscal} = (\text{tmi} + 17.2) / 100$

17.2 is modeled as social contributions.

3.5 Shared valuation revaluation

For non-SCPI real estate strategies:

- $\text{valeur_bien}(y) = \text{prix_achat} * (1 + \text{revalorisation_bien_pct} / 100)^y$

4) Strategy Methodology

4.1 Strategy A - Location Nue

Revenue

Base monthly rent:

- $\text{loyer_nu} = \text{loyer_m2} * \text{surface}$

Year y annual rent:

- $\text{loyer}(y) = \text{loyer_nu} * 12 * (1 + \text{revalorisation_loyer_pct}/100)^{(y-1)} * (1 - \text{vacance_locative_pct}/100)$

Charges

- $\text{charges_copro} = \text{charges_copro_mensuelle} * 12$
- $\text{tf} = \text{taxe_fonciere_mensuelle} * 12$
- $\text{gestion} = \text{loyer}(y) * \text{gestion_locative_pct} / 100$

Tax base and tax

- $\text{revenu_foncier} = \text{loyer}(y) - \text{interets}(y) - \text{charges_copro} - \text{tf} - \text{gestion}$

If $\text{revenu_foncier} > 0$:

- $\text{impot} = \text{revenu_foncier} * \text{taux_fiscal}$

Else:

- $\text{impot} = -\min(\text{abs}(\text{revenu_foncier}), 10700) * \text{tmi} / 100$

Negative impot here models a tax benefit from deductible deficit (simplified).

Cash flow

- $\text{mensualite_an}(y) = \text{mensualite} * 12$ if $y \leq \text{duree_pret_ans}$ else 0
- $\text{cash_flow}(y) = \text{loyer}(y) - \text{mensualite_an}(y) - \text{charges_copro} - \text{tf} - \text{gestion} - \text{impot}$

Patrimoine net

- $\text{patrimoine_net}(y) = \text{valeur_bien}(y) - \text{capital_restant}(y)$

4.2 Strategy B - LMNP Meuble

This strategy was updated to include:

- deductible current charges block,
- amortization capping,
- amortization carryforward,
- deficit carryforward over 10 years.

Revenue

- $\text{loyer_meuble_mensuel} = \text{loyer_nu} * 1.15$
- $\text{loyer}(y) = \text{loyer_meuble_mensuel} * 12 * (1 + \text{revalorisation_loyer_pct}/100)^{(y-1)} * (1 - \text{vacance_locative_pct}/100)$

Amortization hypotheses

- $\text{meublier} = \text{prix_achat} * 0.12$
- $\text{amort_bien_an} = \text{prix_achat} * 0.80 / 30$
- $\text{amort_mob}(y) = \text{meublier} / 7$ for years 1..7, else 0
- $\text{amort_frais_acq_annuel} = \text{input parameter}$

Step-by-step fiscal pipeline

1. Deductible charges excluding amortization

- $\text{charges_courantes}(y) = \text{interets}(y)$
 - $\text{charges_copro}(y)$
 - $\text{tf}(y)$
 - $\text{gestion}(y)$
 - travaux_annuel

- assurance_pno_annuel
- cfe_annuel
- honoraires_comptable_annuel

2. BIC before amortization

- $\text{bic_avant_amort}(y) = \text{loyer}(y) - \text{charges_courantes}(y)$

3. Capped amortization deduction

- $\text{amort_total}(y) = \text{amort_bien_an}$
 - $\text{amort_mob}(y)$
 - $\text{amort_frais_acq_annuel}$
 - $\text{amort_reporte_anterieur}$
- $\text{amort_deductible}(y) = \min(\text{amort_total}(y), \max(0, \text{bic_avant_amort}(y)))$
- $\text{amort_reporte_nouveau} = \text{amort_total}(y) - \text{amort_deductible}(y)$

4. BIC after amortization

- $\text{bic_apres_amort}(y) = \text{bic_avant_amort}(y) - \text{amort_deductible}(y)$

5. Deficit carryforward (10-year stock)

The model keeps yearly deficit buckets with 10-year remaining life.

- If $\text{bic_apres_amort}(y) > 0$:
 - apply deficits from existing buckets (oldest first)
 - $\text{imputation_deficit}(y) = \text{applied amount}$
- If $\text{bic_apres_amort}(y) < 0$:
 - create a new deficit bucket with:
 - $\text{montant} = \text{abs}(\text{bic_apres_amort}(y))$
 - $\text{annees_restantes} = 10$
- $\text{bic_net_imposable}(y) = \max(0, \text{bic_apres_amort}(y) - \text{imputation_deficit}(y))$

6. Tax

- $\text{impot}(y) = \text{bic_net_imposable}(y) * \text{taux_fiscal}$

Cash flow (economic)

LMNP cash flow remains an economic cash flow:

- $\text{cash_flow}(y) = \text{loyer}(y) - \text{mensualite_an}(y) - \text{charges_copro}(y) - \text{tf}(y) - \text{gestion}(y) - \text{impot}(y)$

Important:

- non-cash accounting items (amortissement) do not leave cash,

- but they affect tax through `bic_net_imposable`.

Yearly outputs used in UI

- `charges` = `charges_courantes`
- `amortissement` = `amort_deductible`
- `bic_imposable` = `bic_net_imposable`

4.3 Strategy C - Location Courte Duree (Airbnb-like)

Revenue hypotheses

- `revenu_nuitee` = `loyer_nu` * 2.8
- `taux_occupation` = 120 / 365 (~32.9%) — Paris-like regulatory cap of 120 rental days/year for short-term letting of a primary residence
- `revenu_mensuel` = `revenu_nuitee` * `taux_occupation`
- `revenu(y)` = `revenu_mensuel` * 12 * (1 + `revalorisation_loyer_pct`/100)^(y-1)

Cost hypotheses

- `charges_copro` = `charges_copro_mensuelle` * 12
- `tf` = `taxe_fonciere_mensuelle` * 12
- `plateforme` = `revenu(y)` * 15%
- `conciergerie` = `revenu(y)` * 20%
- `menage` = 150 * 12

Fiscal base

- `mobilier` = `prix_achat` * 0.12 (unified with the LMNP furniture allowance, see 4.2)
- `amort_bien_an` = `prix_achat` * 0.80 / 30
- `amort_mob` = `mobilier` / 7 for years 1..7, else 0
- `bic` = `revenu(y)` - `interets(y)` - `charges_copro` - `tf` - `plateforme` - `conciergerie` - `menage` - `amort_bien_an` - `amort_mob`
- `impot` = max(0, `bic`) * `taux_fiscal`

Cash flow

- `cash_flow(y)` = `revenu(y)` - `mensualite_an(y)` - `charges_copro` - `tf` - `plateforme` - `conciergerie` - `menage` - `impot`

Patrimoine

- `patrimoine_net(y)` = `valeur_bien(y)` - `capital_restant(y)`

4.4 Strategy D - SCPI

Capital and return hypotheses

- $\text{frais_entree} = 9\%$
- $\text{capital_net} = \text{apport} * (1 - 0.09)$
- $\text{rendement} = \text{rendement_scpi} / 100$
- $\text{revalorisation_parts} = 1\%/year$

Yearly loop

- $\text{revenu}(y) = \text{valeur_parts}(y-1) * \text{rendement}$
- $\text{impot}(y) = \text{revenu}(y) * \text{taux_fiscal}$
- $\text{cash_flow}(y) = \text{revenu}(y) - \text{impot}(y)$
- $\text{valeur_parts}(y) = \text{valeur_parts}(y-1) * (1 + 0.01)$

In SCPI rows:

- $\text{patrimoine_net} = \text{valeur_parts}$

4.5 Strategy E - Residence Principale

Economic benefit approach

No rental income is modeled. Instead, the model uses avoided rent:

- $\text{loyer_economise}(y) = \text{loyer_nu} * 12 * (1 + \text{revalorisation_loyer_pct}/100)^{(y-1)}$

Costs and cash flow differential

- $\text{charges} = \text{charges_copro} + \text{tf}$
- $\text{cash_flow}(y) = \text{loyer_economise}(y) - \text{mensualite_an}(y) - \text{charges_copro} - \text{tf}$

Patrimoine

- $\text{patrimoine_net}(y) = \text{valeur_bien}(y) - \text{capital_restant}(y)$

5) Output Indicators - Definitions

5.1 Global summary indicators

- prix_achat
- frais_notaire
- cout_total
- emprunt
- mensualite
- $\text{loyer_mensuel_brut}$
- $\text{rendement_brut_nu} = (\text{loyer_nu} * 12 / \text{prix_achat}) * 100$

5.2 Yearly table indicators

- annee : year index 1..horizon
- loyer_annuel or loyer_economise
- interets : annual interest
- charges : annual expenses block (strategy-dependent)

- mensualite_an: annual debt service while loan is active
- amortissement: LMNP deductible amortization in current year
- impot: annual modeled tax
- cash_flow: annual economic cash flow
- cash_flow/mois (UI): $\text{round}(\text{cash_flow} / 12)$
- valeur_bien (except SCPI)
- capital_restant (except SCPI)
- patrimoine_net

5.3 Strategy-level synthetic indicators

- cash_flow_moyen
 - average monthly cash flow over horizon
 - implementation: $\text{round}(\text{sum}(\text{cash_flow annual rows}) / \text{horizon} / 12)$
- patrimoine_net_final
 - last yearly patrimoine_net
- rendement_brut
 - Location Nue: $(\text{loyer_nu} * 12 / \text{prix_achat}) * 100$
 - LMNP: $(\text{loyer_meuble} * 12 / \text{prix_achat}) * 100$
 - Courte Duree: $(\text{revenu_mensuel} * 12 / \text{prix_achat}) * 100$
 - SCPI: rendement_scpi input
 - Residence Principale: 0
- tri (IRR)
 - computed from yearly cash flow series + terminal value component

6) TRI / IRR Construction

6.1 General numerical method

- Newton-Raphson iterations
- max iterations: 1000
- tolerance: $1e-8$
- return accepted only if rate in a bounded interval (model guardrail)

6.2 Cash-flow vectors by strategy

Location Nue:

- initial outflow: $-(\text{apport} + \text{frais_notaire})$
- yearly inflows: annual cash_flow
- final add-on: $\text{valeur_fin} - \text{capital_restant_fin}$

LMNP:

- initial outflow: $-(\text{apport} + \text{frais_notaire} + \text{mobilier})$
- yearly inflows: annual cash_flow
- final add-on: $\text{valeur_fin} - \text{capital_restant_fin}$

Courte Duree:

- initial outflow: $-(\text{apport} + \text{frais_notaire} + \text{mobilier})$
- yearly inflows: annual cash_flow
- final add-on: $\text{valeur_fin} - \text{capital_restant_fin}$

SCPI:

- initial outflow: -apport
- yearly inflows: annual cash_flow
- final add-on: final valeur_parts

Residence Principale:

- initial outflow: $-(\text{apport} + \text{frais_notaire})$
- yearly inflows: annual cash_flow
- final add-on: valeur_fin

7) Rounding and Display Rules

- Most yearly values are rounded to nearest EUR in backend outputs.
- UI cash flow/mois is derived from annual cash flow and then rounded.
- Small annual differences may appear as 0 or +/-1 per month after rounding.

8) Modeling Hypotheses and Simplifications

The model intentionally simplifies several real-life topics.

Included assumptions:

- deterministic yearly growth rates,
- deterministic vacancy and management rates,
- deterministic short-term rental multipliers and fee rates,
- fixed SCPI fee and revaluation assumptions,
- simplified deficit and amortization carry mechanisms.

Not fully modeled:

- resale transaction costs and taxation details (except current simplified logic),
- loan insurance and banking ancillary costs,
- maintenance CAPEX lifecycle details,
- legal constraints and regime edge cases,
- inflation by cost line,
- stochastic scenarios (probability distributions, stress cases).

9) Consistency Notes

- The methodology document, backend formulas, UI tables, and PDF export should remain aligned.
- If backend logic changes, this file must be updated in the same change set.